UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 4: Matrix Operations in Excel

By

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Objectives

- To classify types of matrix operations in Excel.
- To conduct all matrix operations in Excel.

Original notes by NFM Noor

Matrix Operations

- Excel provides tool for solving matrix equations.

- Matrix operations:
  1) Addition
  2) Subtraction
  3) Scalar Multiplication
  4) Matrix Multiplication
  5) Transpose
  6) Determinant
  7) Inverse

Original notes by NFM Noor

Example 1

Given the following matrices:

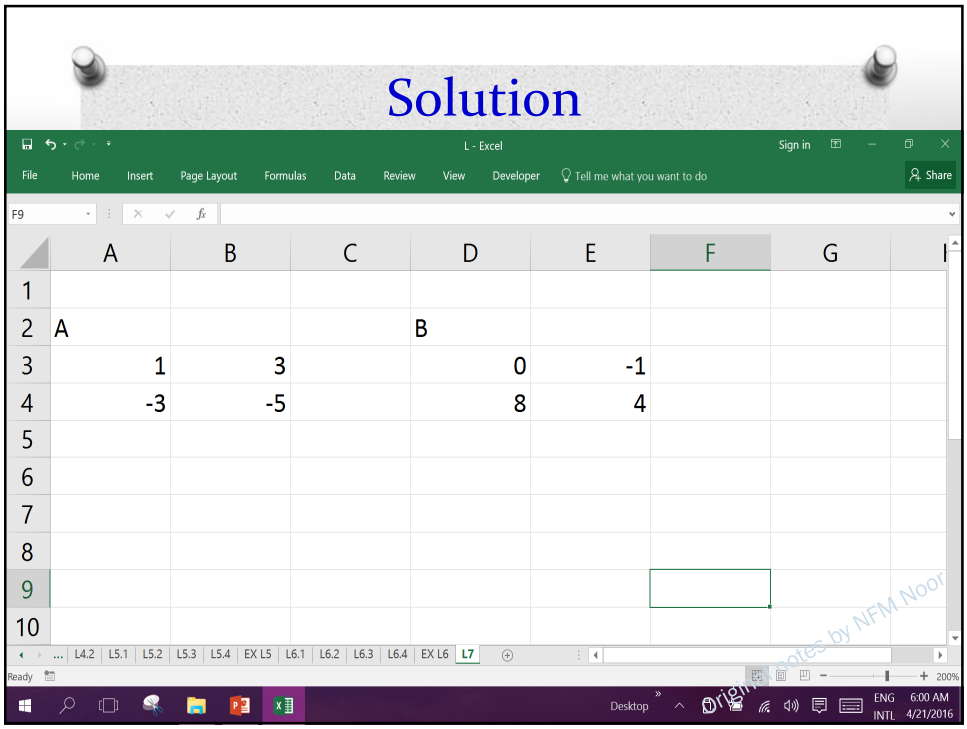
$$A = \begin{pmatrix} 1 & 3 \\ -3 & -5 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & -1 \\ 8 & 4 \end{pmatrix}$$

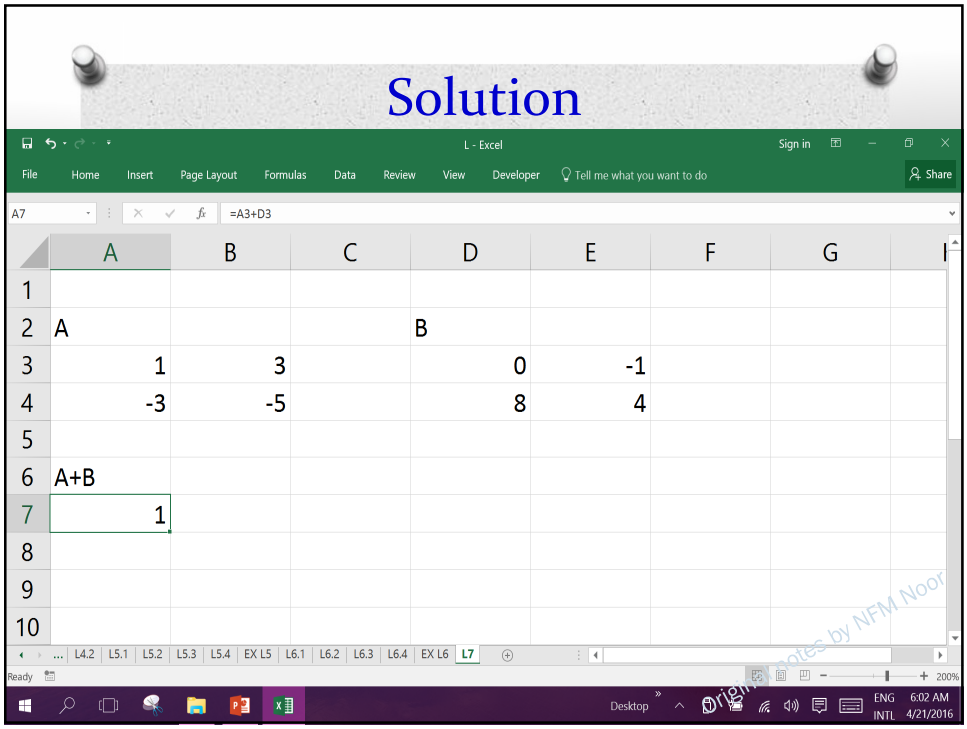
Find $A + B, A - B, -5A$ and BA.

Original notes by NFM Noor

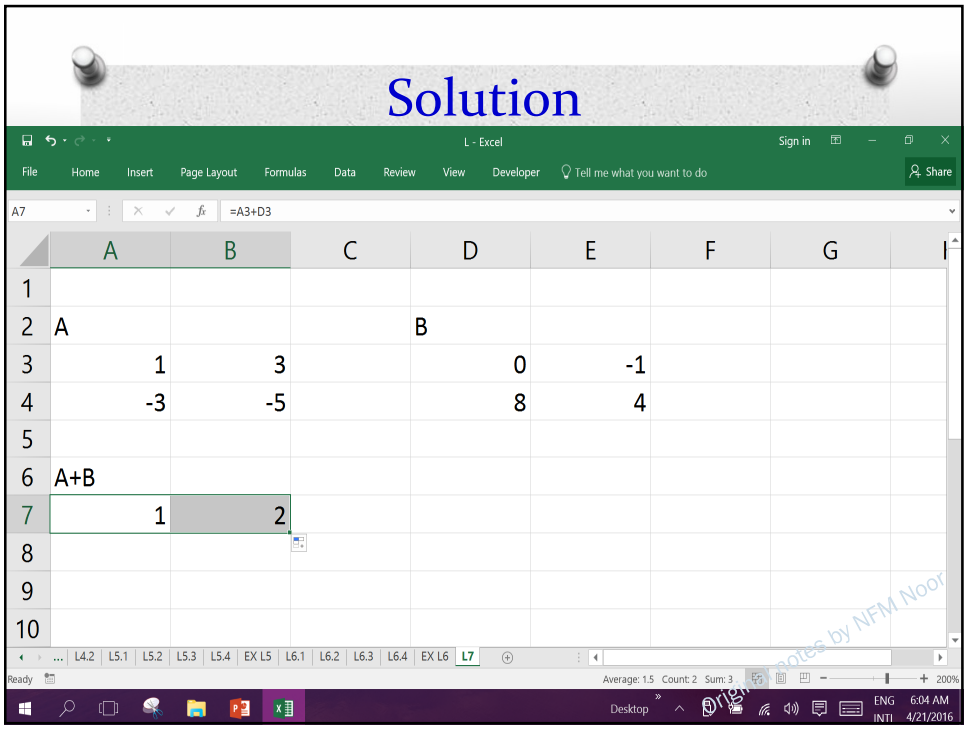
Solution



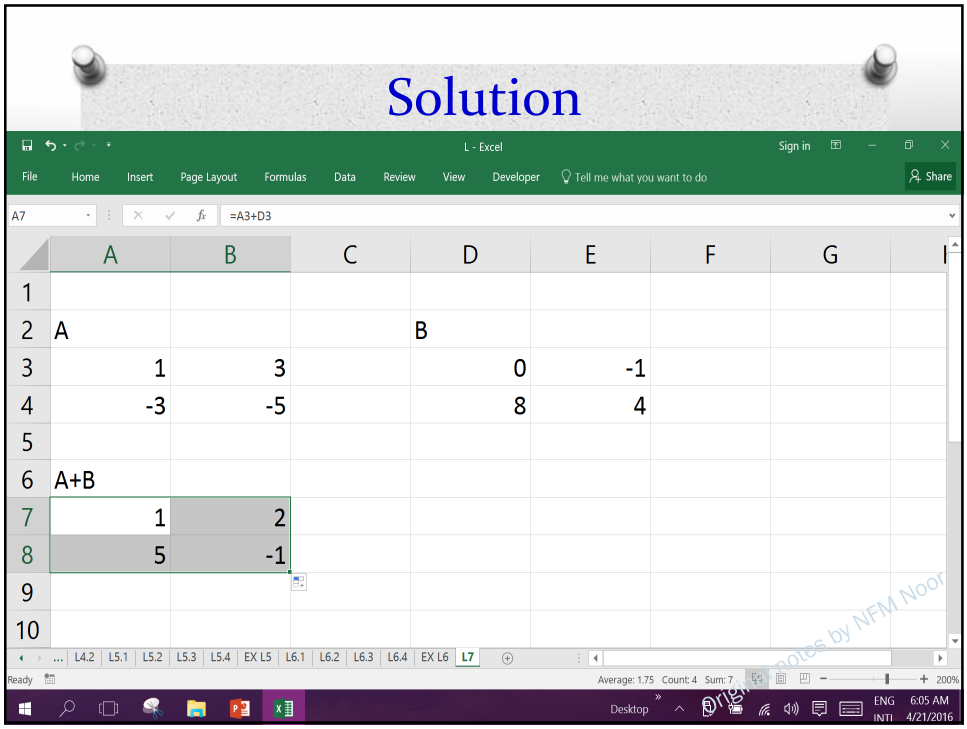
Solution



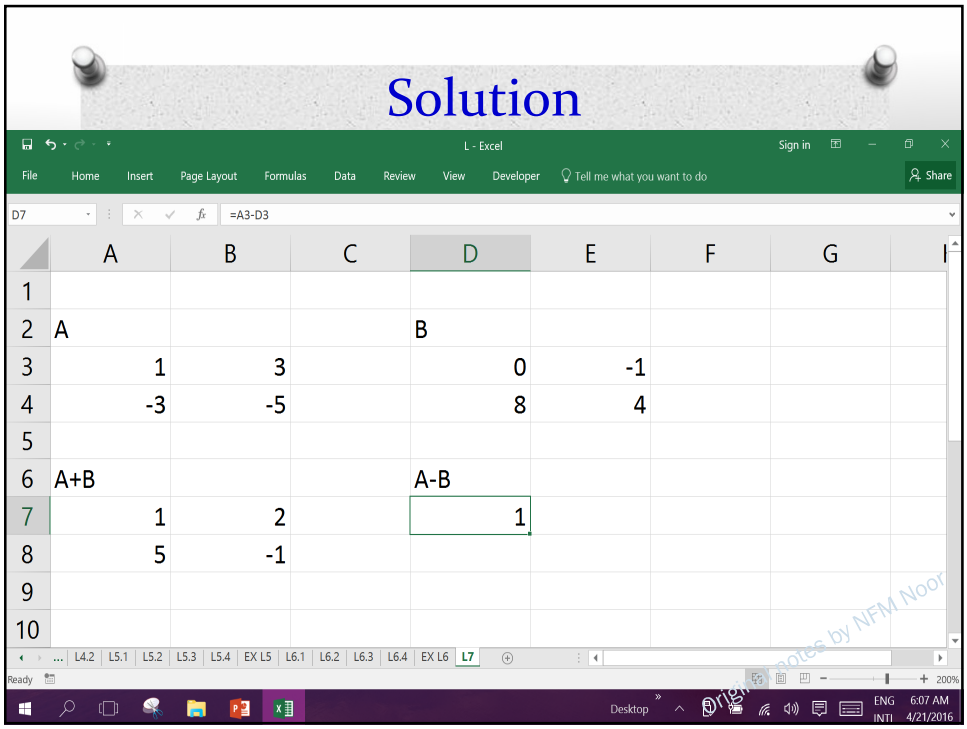
Solution



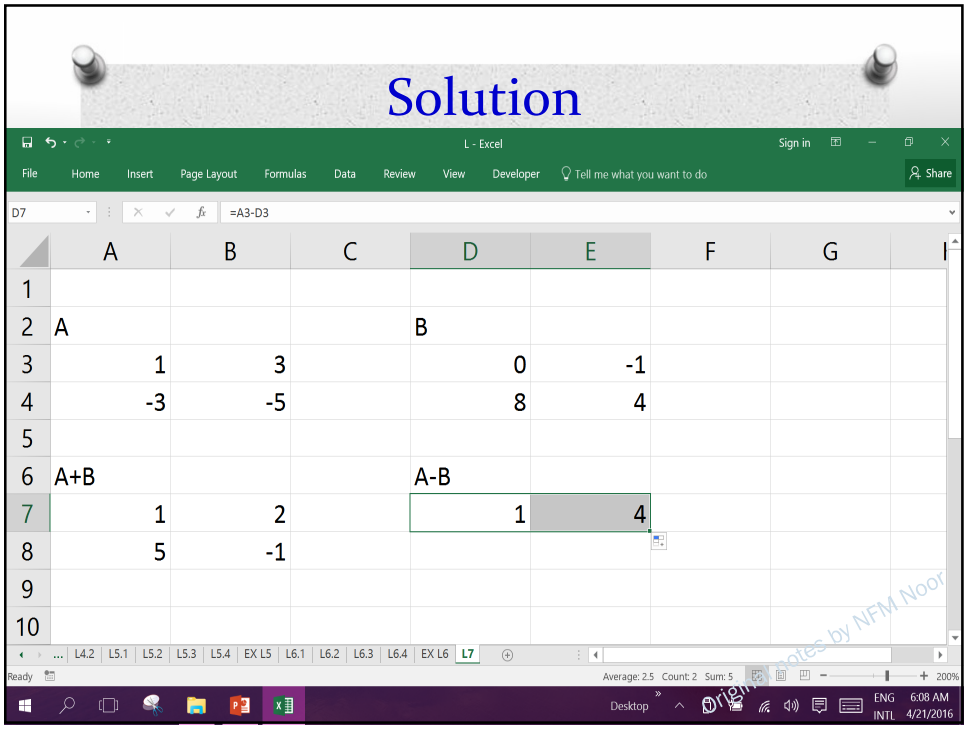
Solution



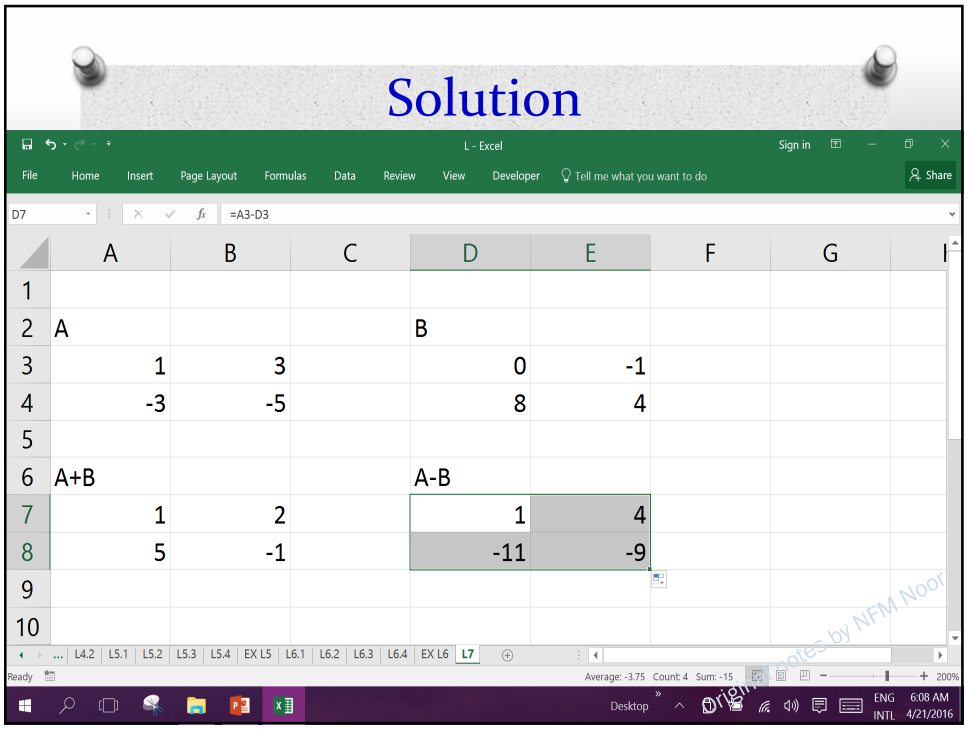
Solution



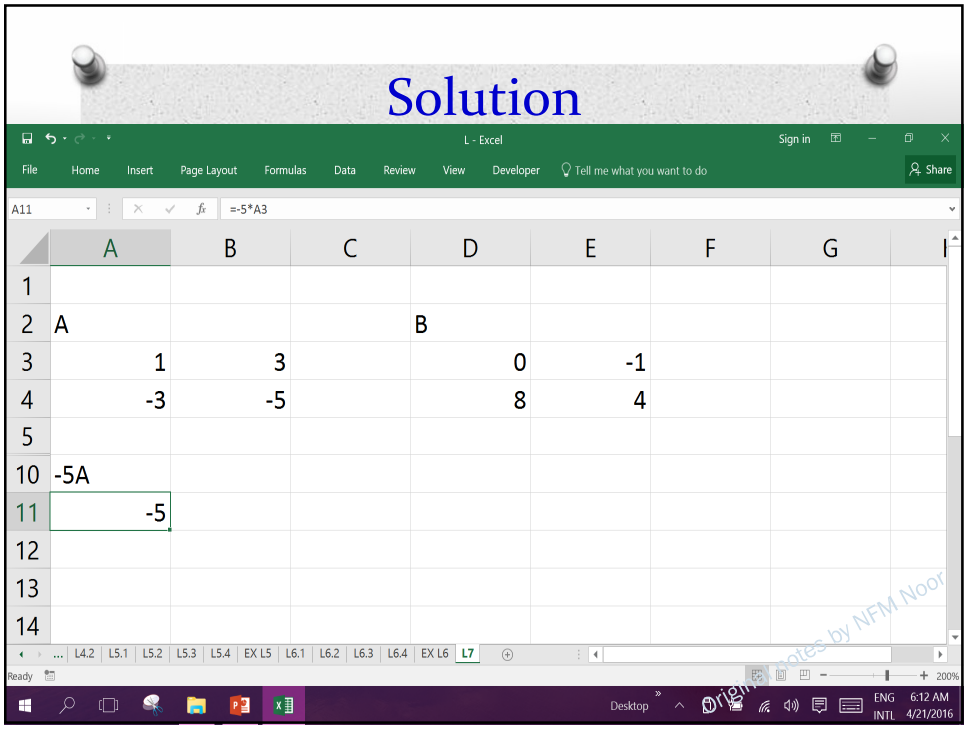
Solution



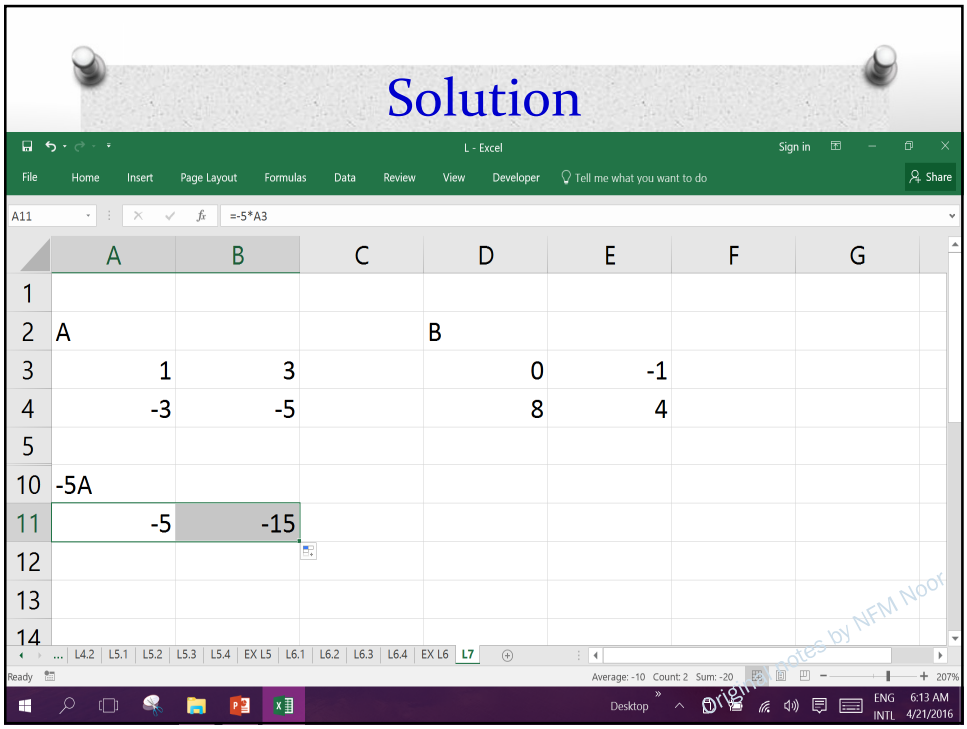
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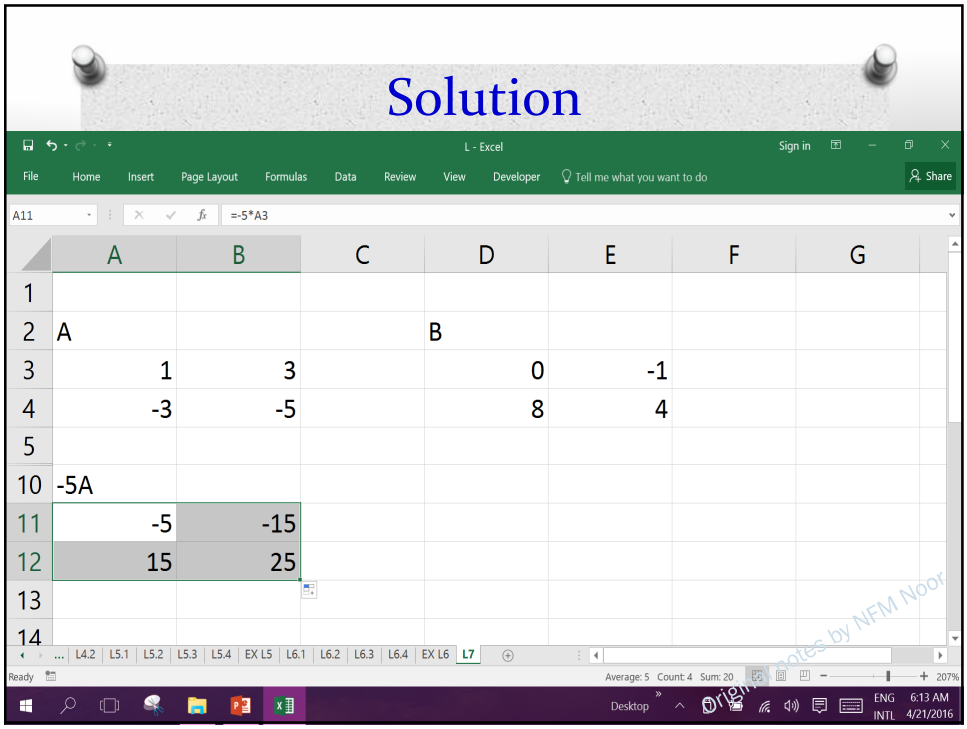
Solution



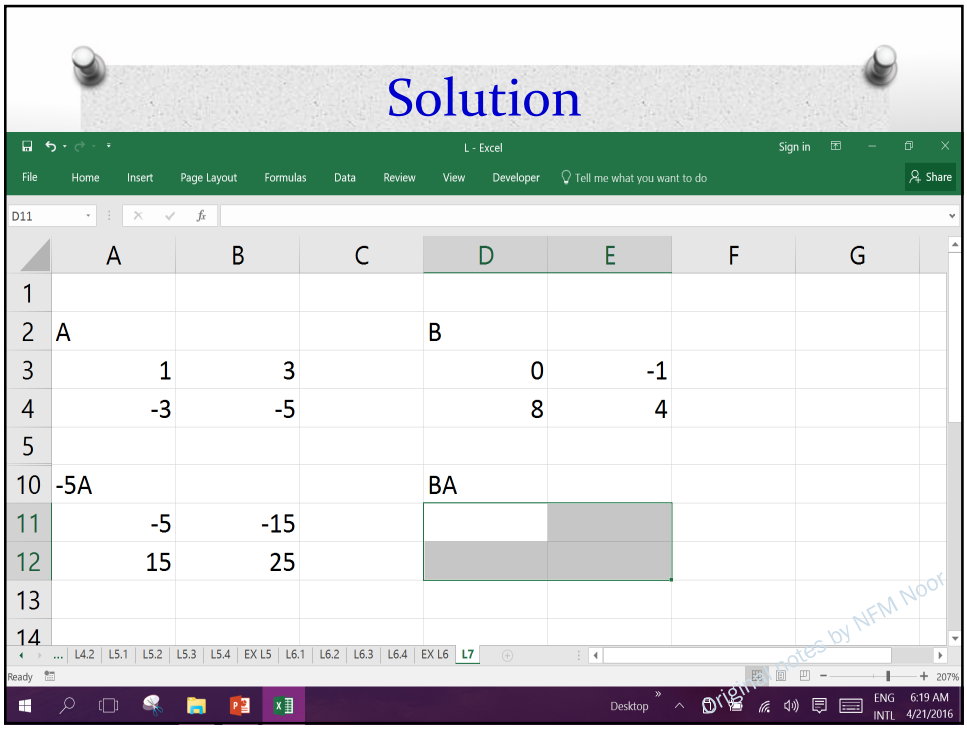
Solution



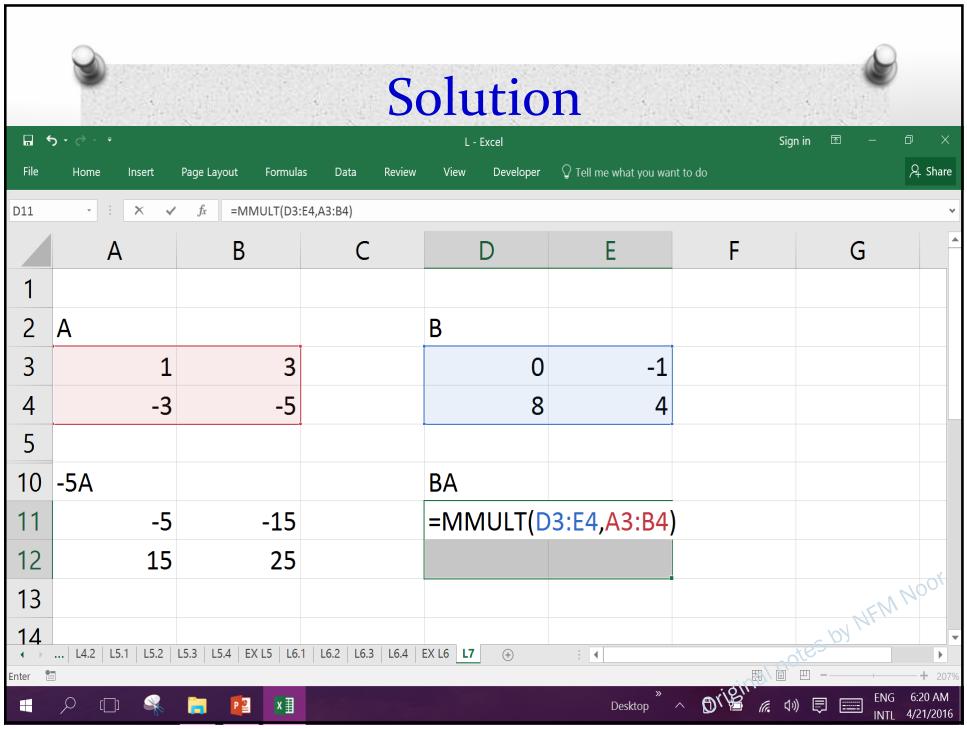
Solution



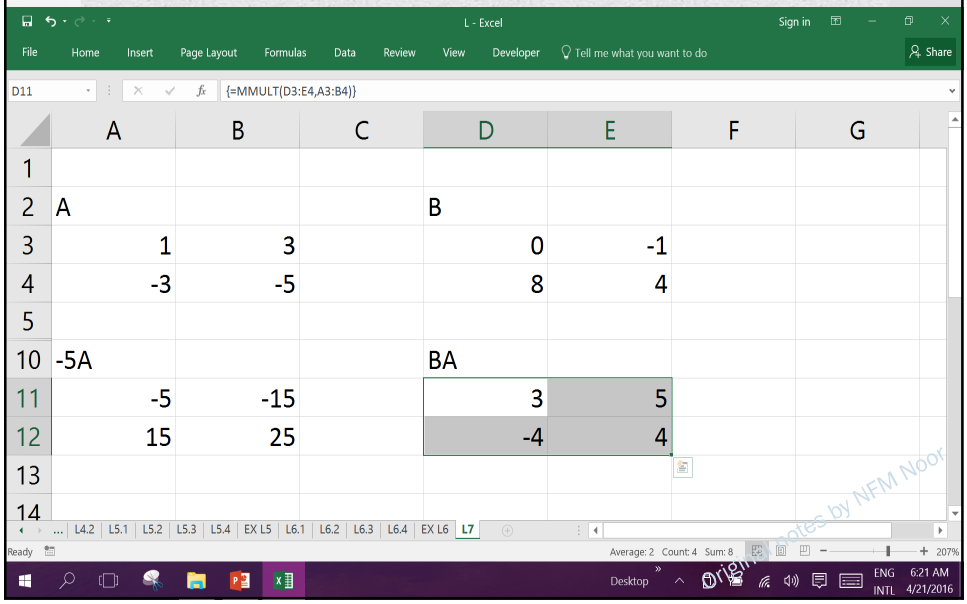
Solution



Solution



Solution



Exercise 1

Given the following matrices:

$$A = \begin{pmatrix} 4 & -5 & 0 \\ -1 & -2 & 9 \\ 3 & 3 & 8 \end{pmatrix}, \mathrm{B} = \begin{pmatrix} -7 & 1 & 2 \\ 10 & 0 & -4 \\ 3 & -5 & -1 \end{pmatrix}$$

Find $A + B, B - 3A, -\frac{1}{2}B$ and $4AB$.

Original notes by NFM Noor

Example 2

Given the following matrices:

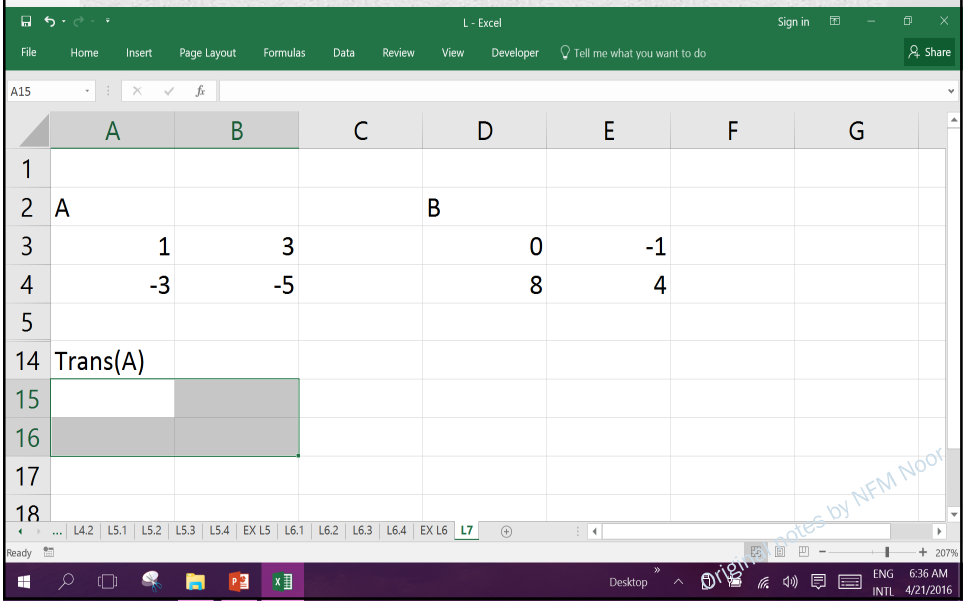
$$A = \begin{pmatrix} 1 & 3 \\ -3 & -5 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & -1 \\ 8 & 4 \end{pmatrix}$$

Find A^T and B^T.

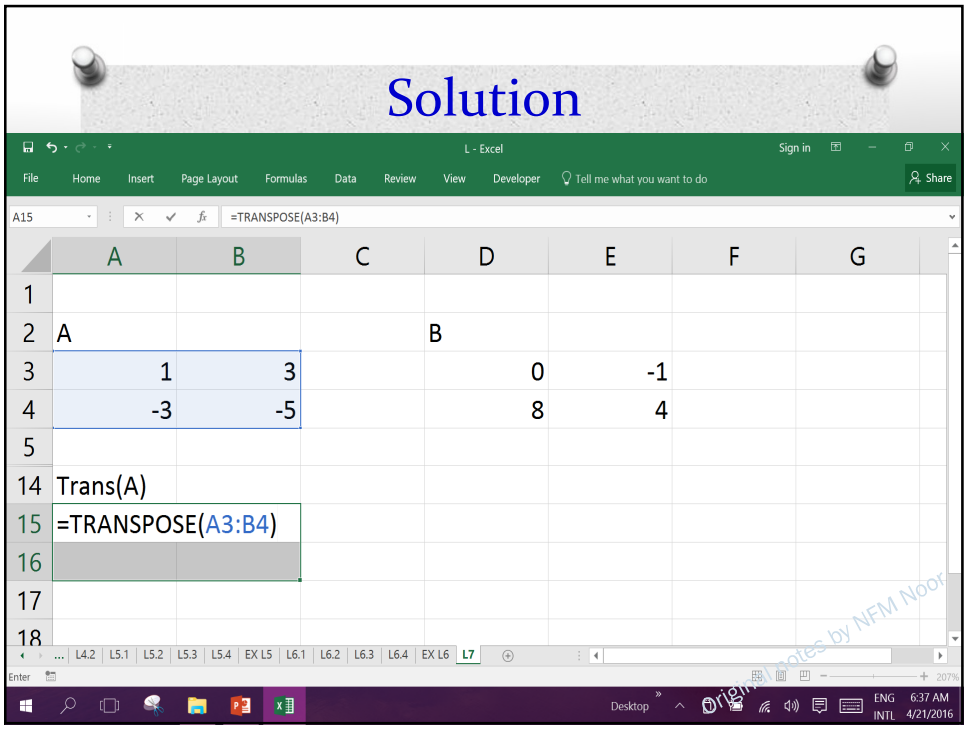
Original notes by NFM Noor

Solution

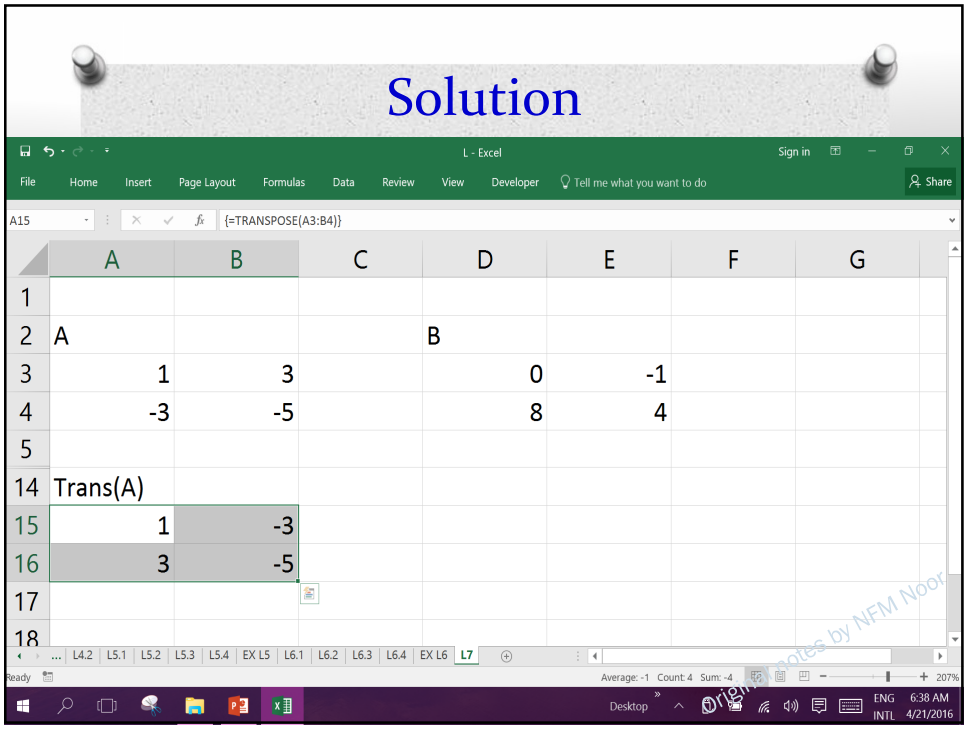


Original notes by NFM Noor

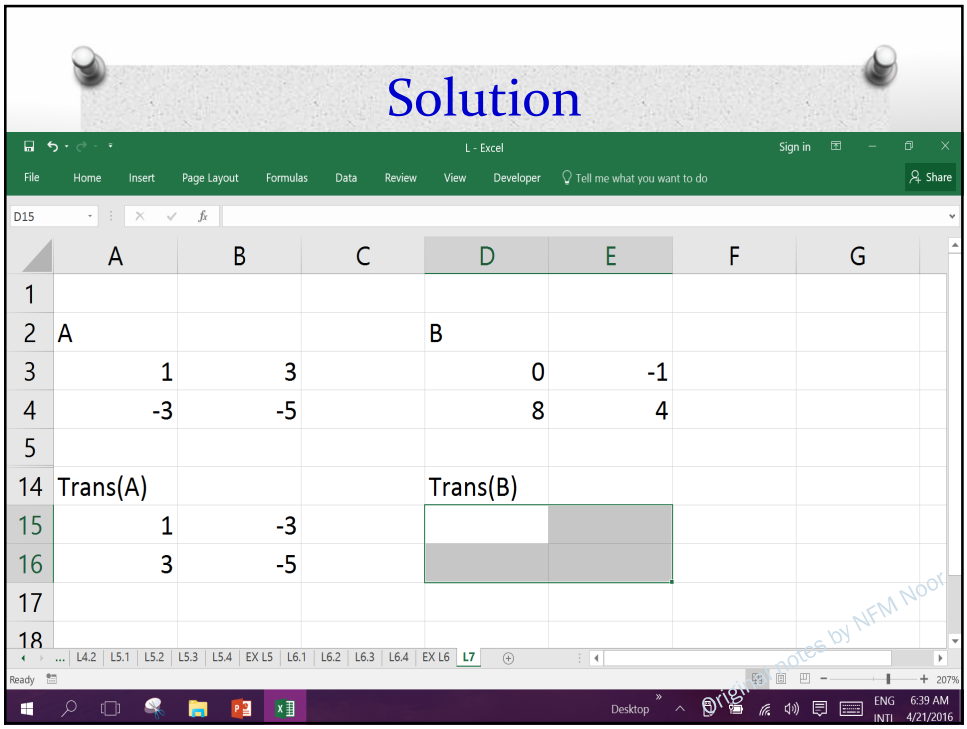
Solution



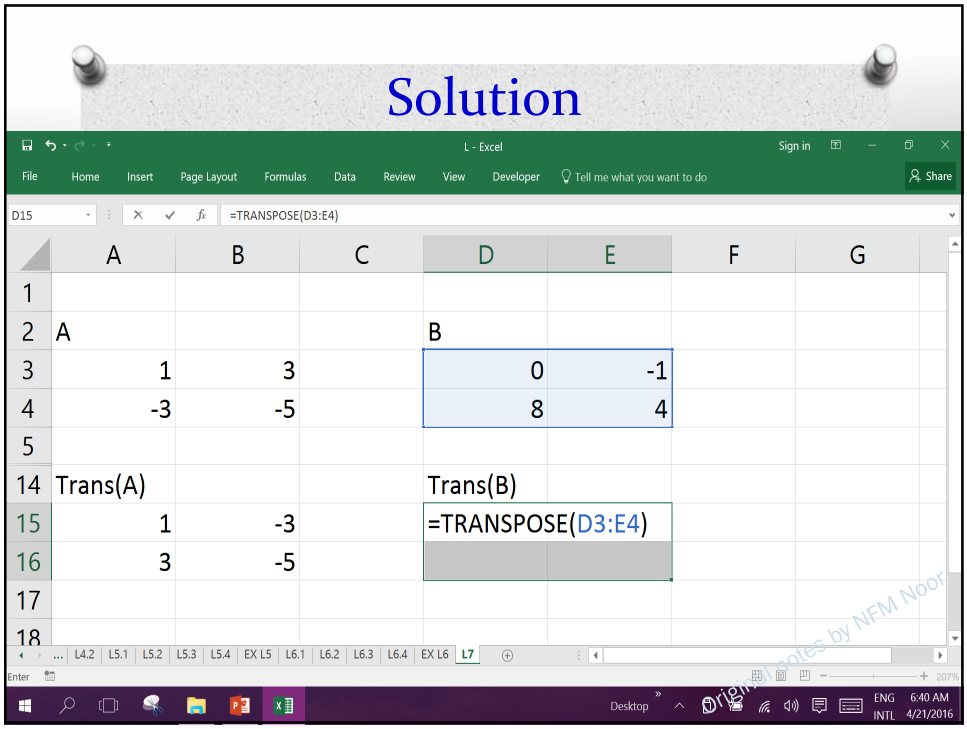
Solution



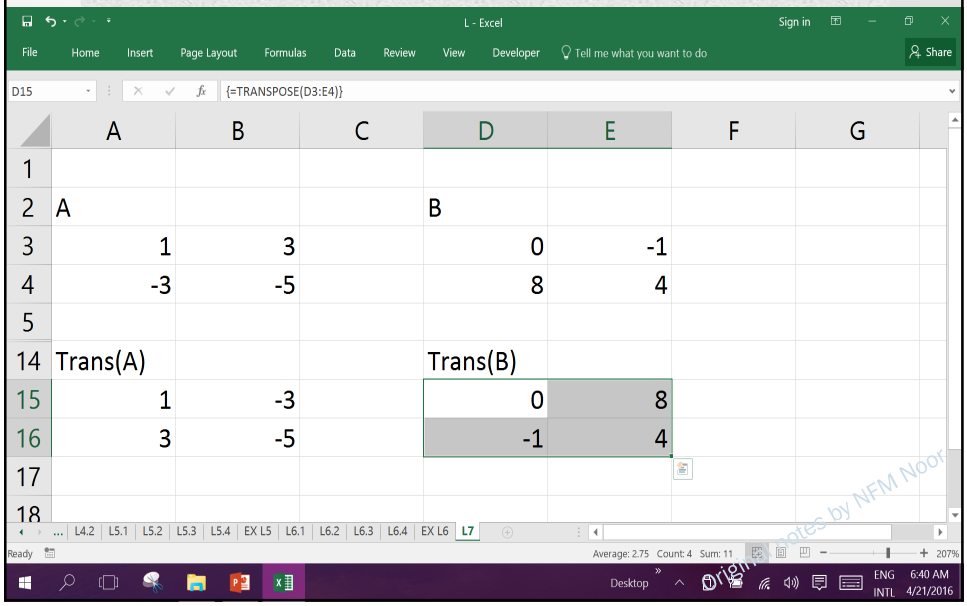
Solution



Solution



Solution



Exercise 2

Given the following matrices:

$$A = \begin{pmatrix} 4 & -5 & 0 \\ -1 & -2 & 9 \\ 3 & 3 & 8 \end{pmatrix}, \mathrm{B} = \begin{pmatrix} -7 & 1 & 2 \\ 10 & 0 & -4 \\ 3 & -5 & -1 \end{pmatrix}$$

Find $-3A^T + B^T$.

Original notes by NFM Noor

Example 3

Given the following matrices:

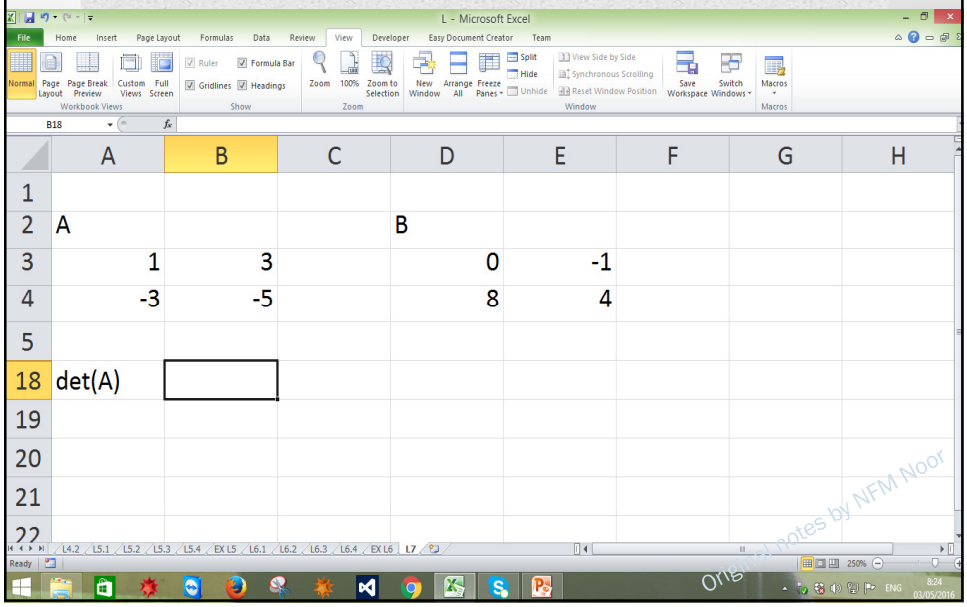
$$A = \begin{pmatrix} 1 & 3 \\ -3 & -5 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & -1 \\ 8 & 4 \end{pmatrix}$$

Find $\det(A)$ and $\det(B)$.

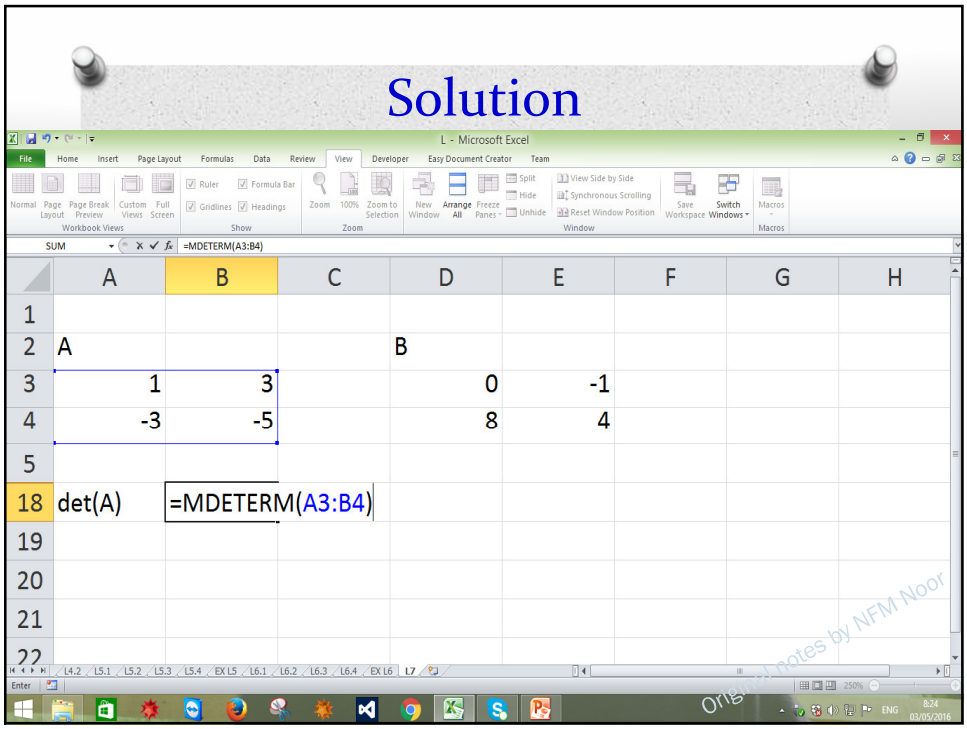
Original notes by NFM Noor

Solution

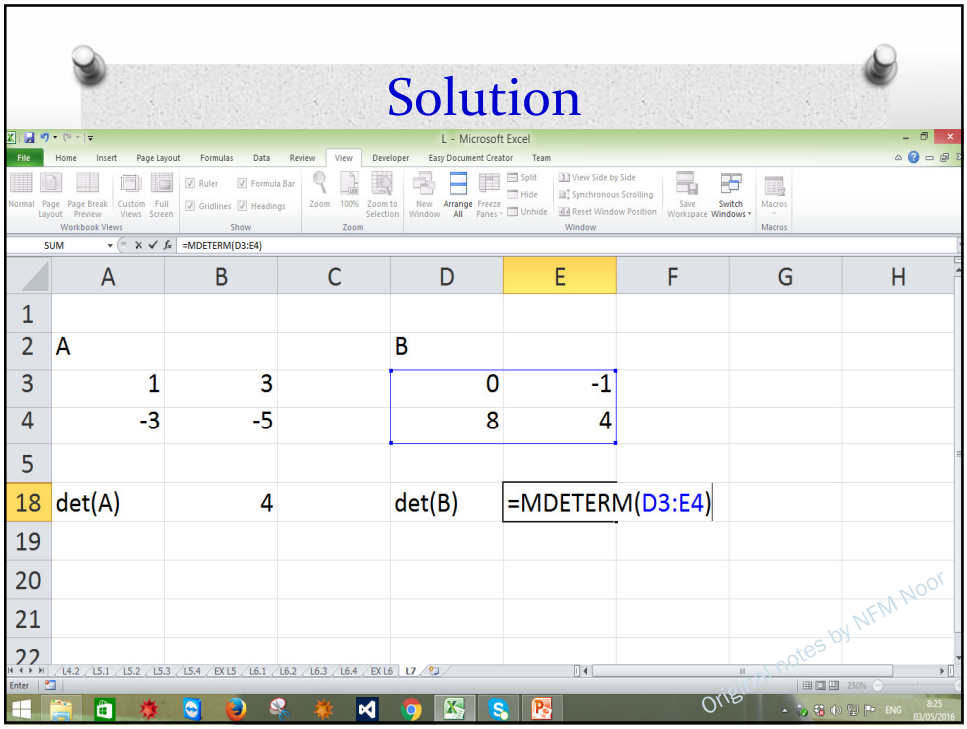


Original notes by NFM Noor

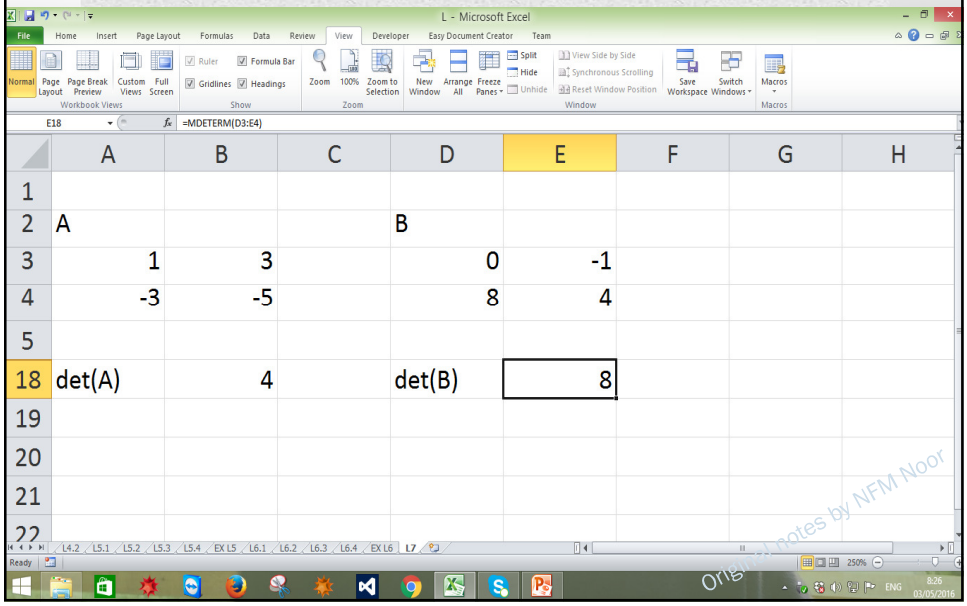
Solution



Solution



Solution



Exercise 3

Given the following matrices:

$$A = \begin{pmatrix} 4 & -5 & 0 \\ -1 & -2 & 9 \\ 3 & 3 & 8 \end{pmatrix}, \mathrm{B} = \begin{pmatrix} -7 & 1 & 2 \\ 10 & 0 & -4 \\ 3 & -5 & -1 \end{pmatrix}$$

Find $\det(A), \det(B), \det(AB)$ and $\det(BA)$.

Original notes by NFM Noor

Example 4

Given the following matrices:

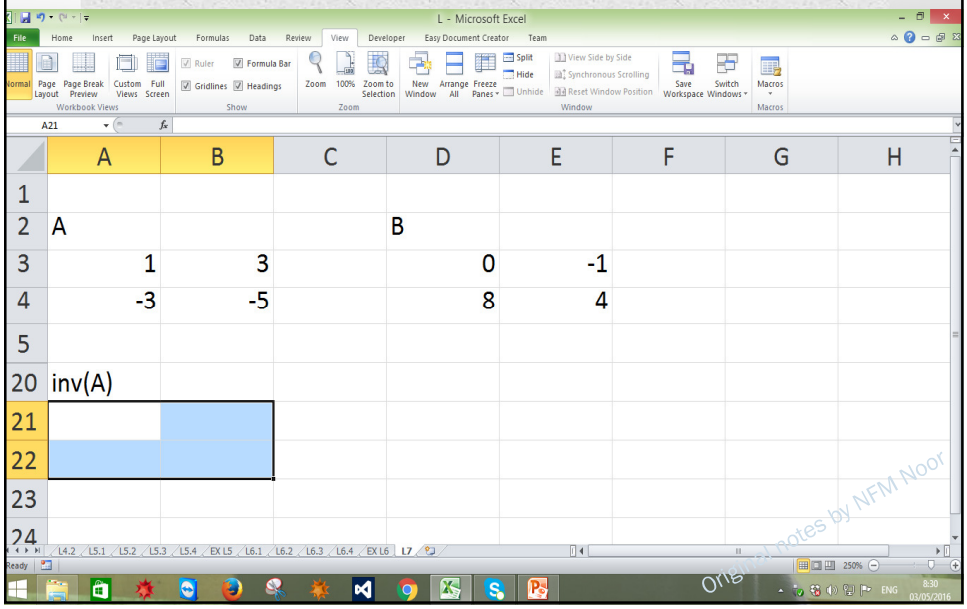
$$A = \begin{pmatrix} 1 & 3 \\ -3 & -5 \end{pmatrix}$$

$$B = \begin{pmatrix} 0 & -1 \\ 8 & 4 \end{pmatrix}$$

Find A^{-1} and B^{-1}.

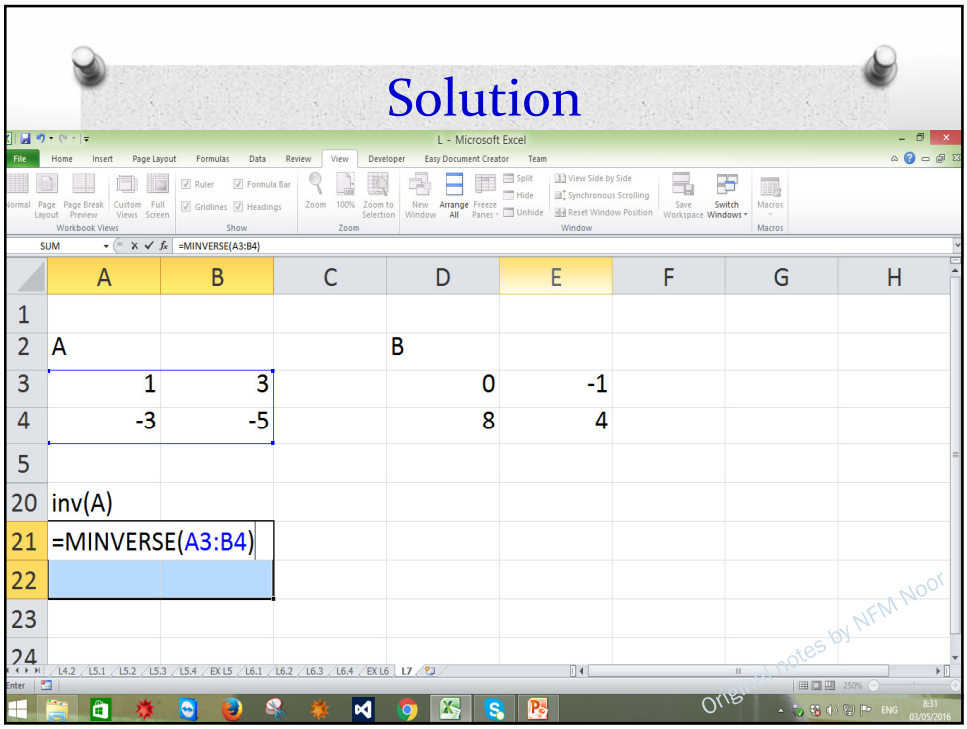
Original notes by NFM Noor

Solution

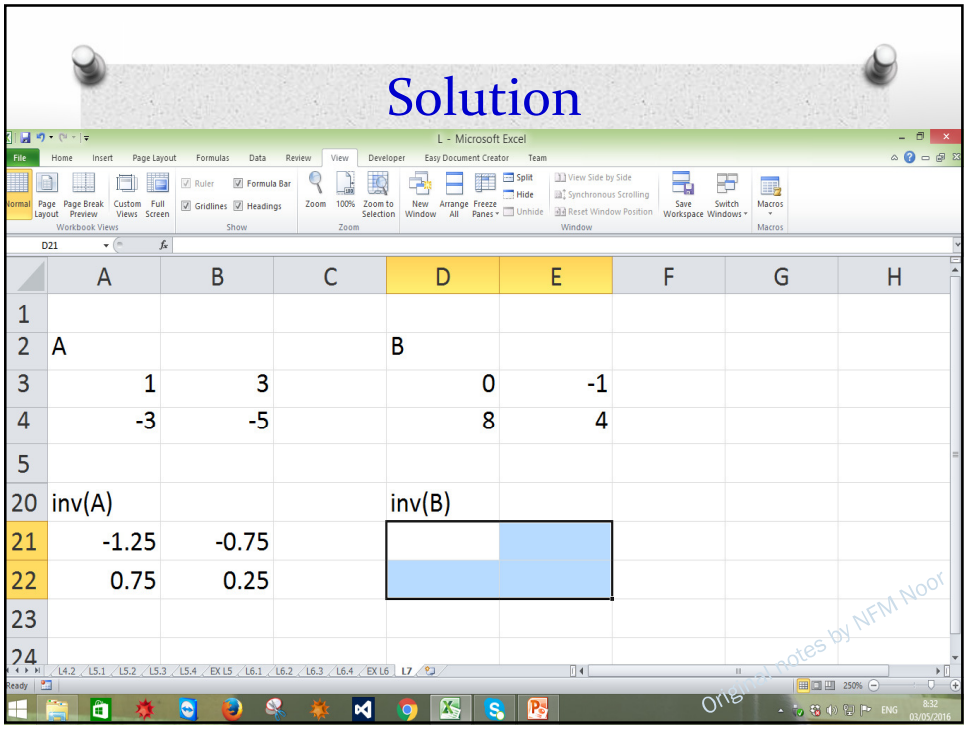


Original notes by NFM Noor

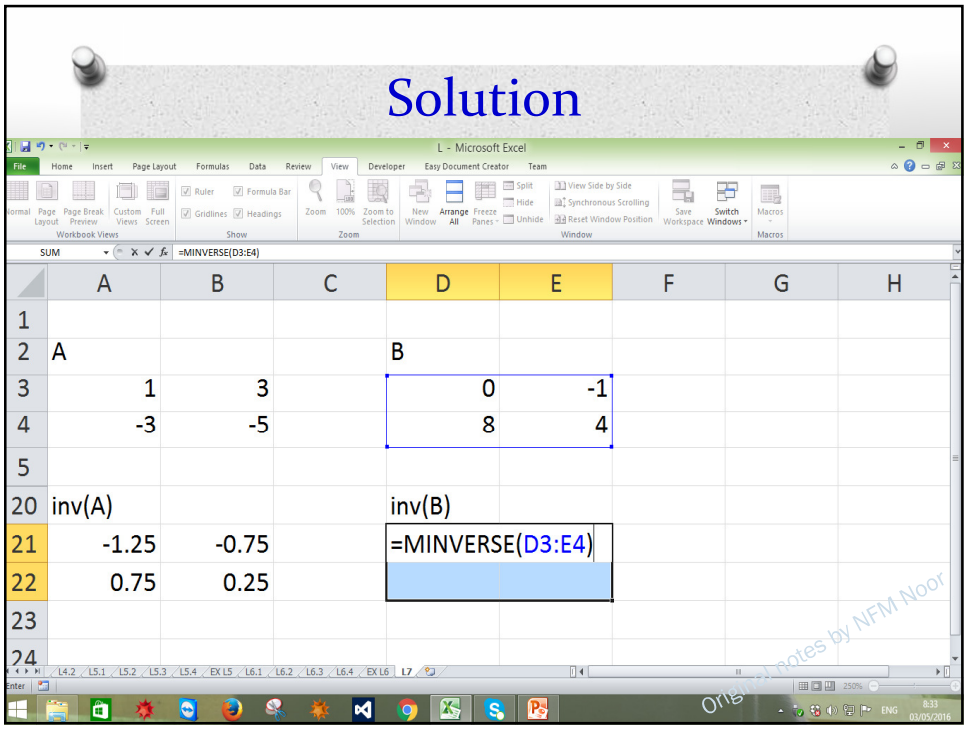
Solution



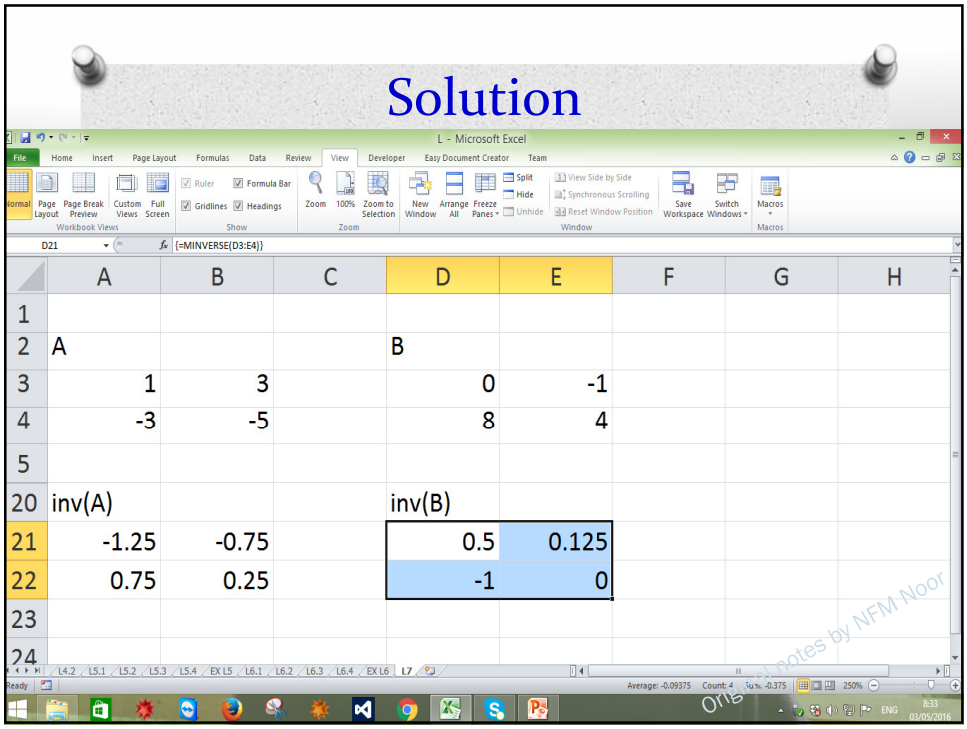
Solution



Solution



Solution



Exercise 4

Given the following matrices:

$$A = \begin{pmatrix} 4 & -5 & 0 \\ -1 & -2 & 9 \\ 3 & 3 & 8 \end{pmatrix}, \text{B} = \begin{pmatrix} -7 & 1 & 2 \\ 10 & 0 & -4 \\ 3 & -5 & -1 \end{pmatrix}$$

Find $A^{-1}, B^{-1}, (AB)^{-1}$ and $(BA)^{-1}$.

Original notes by NFM Noor

Summary

- Matrix operations can be conducted easily in Excel.
- Basic matrix operations are addition, subtraction, scalar multiplication and matrix multiplication.
- Excel built-in functions are used to find matrix transpose, determinant and inverse.

Original notes by NFM Noor